

SAMJIYON, Korea, Democratic People's Republic of

WMO: 470050

Lat: 41.817N

Lon: 128.300E

Elev: 4531

StdP: 12.44

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.2	-17.2	-25.1	1.6	-19.8	-21.3	2.0	-15.9	14.5	3.0	12.2	2.1	2.8	180	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	77.3	64.6	74.7	63.2	72.0	62.0	67.6	74.2	66.1	71.5	64.6	69.1	4.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.6	112.0	70.5	64.3	106.8	68.9	63.0	102.1	67.6	35.0	74.4	33.6	71.8	32.3	69.1	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.0	10.9	9.2	-28.7	83.4	5.2	1.9	-32.4	84.8	-35.4	85.9	-38.3	87.0	-42.0	88.4		
(5)			-29.2	71.0	5.0	3.1	-32.8	73.2	-35.8	75.0	-38.6	76.8	-42.3	79.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.6	1.6	6.1	17.7	34.3	47.9	56.2	63.0	61.1	50.3	38.0	19.6	5.9		
	DBStd	23.20	8.87	9.82	10.09	9.39	7.43	5.36	4.50	5.63	6.56	8.18	11.07	9.91		
	HDD50	7077	1500	1229	1000	474	129	11	0	3	75	377	911	1368		
	HDD65	11503	1965	1649	1465	920	532	265	92	144	440	837	1361	1833		
	CDD50	1104	0	0	0	4	63	198	403	347	85	4	0	0		
	CDD65	56	0	0	0	0	1	2	30	23	0	0	0	0		
	CDH74	313	0	0	0	0	18	35	155	103	2	0	0	0		
	CDH80	27	0	0	0	0	0	0	14	13	0	0	0	0		
	WSAvg	3.4	3.4	3.4	3.6	3.9	3.7	2.7	2.8	2.9	3.1	3.4	3.8	3.8		
	PrecAvg	39.8	0.6	0.7	1.2	2.2	3.9	5.9	9.8	7.4	3.8	1.8	1.9	0.9		
	PrecMax	55.4	2.2	2.8	2.3	4.1	7.6	11.1	16.9	12.5	9.6	3.3	5.2	1.9		
(15) Precipitation	PrecMin	28.1	0.2	0.2	0.5	0.8	1.4	2.0	3.4	2.4	1.1	0.4	0.6	0.2		
	PrecStd	5.9	0.4	0.6	0.5	0.8	1.4	2.3	3.4	2.6	1.9	0.8	1.0	0.4		
	DB	28.0	35.6	50.0	67.1	76.0	77.7	82.1	81.4	72.5	66.2	48.2	34.5			
	MCWB	24.1	30.3	38.3	49.6	57.1	61.7	67.4	67.9	59.0	50.1	37.5	29.6			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	22.9	29.9	42.5	59.6	70.9	74.5	78.0	76.6	69.0	60.3	42.5	28.3			
	2% MCWB	20.2	25.5	34.0	45.3	53.8	59.8	66.0	65.7	57.0	47.8	36.2	25.7			
	5% DB	19.8	26.0	37.7	54.4	66.6	71.3	75.1	73.5	65.9	55.6	38.5	24.4			
	5% MCWB	17.4	22.4	31.1	42.4	52.0	58.6	64.5	64.6	55.9	45.0	33.5	22.0			
	10% DB	16.1	22.2	33.8	49.6	62.2	68.1	72.2	70.6	62.8	51.5	34.9	20.9			
	10% MCWB	14.2	19.3	29.0	39.6	49.7	57.7	63.9	63.2	54.0	42.6	31.6	18.9			
	0.4% WB	24.4	32.6	39.7	51.1	59.3	64.8	70.2	69.8	62.7	51.8	40.5	30.5			
	0.4% MCDB	27.2	35.1	48.7	63.6	69.3	73.4	77.5	77.0	68.4	62.1	45.5	33.5			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	20.5	25.9	34.7	46.2	56.0	62.1	68.1	67.5	59.9	49.0	37.0	25.9			
	2% MCDB	22.6	29.5	40.7	57.9	67.3	70.2	75.1	73.8	65.4	58.6	41.6	28.2			
	5% WB	17.5	22.5	31.9	43.4	53.5	60.4	66.6	66.1	57.8	46.4	34.1	22.1			
	5% MCDB	19.5	25.9	36.8	53.0	63.8	68.1	72.4	71.3	63.3	53.7	37.7	24.1			
	10% WB	14.2	19.4	29.2	40.4	51.0	58.9	65.2	64.7	55.6	43.9	31.6	19.0			
	10% MCDB	15.9	22.2	33.9	48.5	59.9	65.9	70.1	68.8	60.5	49.8	34.6	20.9			
	MDBR	20.1	21.4	20.7	19.2	20.4	19.3	15.7	16.4	20.9	21.1	17.7	18.2			
	5% DB	22.4	22.9	23.4	26.2	28.3	26.2	21.7	20.5	25.0	27.2	19.9	19.1			
(36) Mean Daily Temperature Range	MCWBR	19.6	18.8	16.8	15.4	14.9	14.3	11.3	10.9	14.4	15.9	14.0	16.4			
	MCDBR	21.6	21.4	21.0	24.3	25.2	21.3	17.1	16.0	19.7	24.7	17.9	18.7			
	MCWBR	19.2	17.8	15.9	15.8	14.7	12.9	10.0	9.3	12.5	15.4	13.4	16.5			
	MCDBR	21.6	21.4	21.0	24.3	25.2	21.3	17.1	16.0	19.7	24.7	17.9	18.7			
(40) Clear-Sky Solar Irradiance	taub	0.287	0.338	0.402	0.468	0.501	0.510	0.463	0.420	0.376	0.377	0.332	0.293			
	taud	2.147	2.016	1.910	1.862	1.854	1.906	2.125	2.248	2.303	2.166	2.166	2.151			
	Ebn at Noon	267	267	263	254	249	247	258	266	271	252	246	252			
	Edh at Noon	32	43	55	62	64	61	48	42	36	37	31	29			
(44) All-Sky Solar Radiation	RadAvg	782	1093	1410	1519	1648	1608	1496	1462	1354	1044	709	633			
	RadStd	44	51	81	98	147	158	154	125	125	91	71	33			

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page